Devkishan Khatri

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SUMMARY

Naval Architect / Marine Engineer with 1+ year of industrial experience in floating structure analysis, mooring design, and engineering automation.Graduated from IIT Kharagpur (Dual Degree) with highest departmental GPA. Passionate about applying rigorous simulation and computational methods to solve real-world offshore and subsea engineering challenges.

SKILLS

Softwares: Orcawave, OrcaFlex, Ansys(Fluent, AQWA), Maxsurf, AutoCAD, SolidWorks, Rhino3D, GMSH, MATLAB
Technical Skills: CFD, Hydrodynamics, Mooring Design, FEA, VIV, Floating Structure, C++, Python, VBA, AutoLISP
Standard Codes: API RP 2SK, DNV RP C205, DNV OS E301, IS 875(part 3), IS 811, IS 2911

EXPERIENCE

Naval Architect (R & D) | Quantsolar Technologies Pvt. Ltd., Hyderabad — **May 2025 - present**

Engineering Analysis & Research and Development

- Conducted **CFD simulations** using **ANSYS Fluent** to estimate **wind loads** for mooring analysis of floating solar islands
- Executed **mooring analysis** using **OrcaFlex** under different load cases to ensure safe station keeping in various condition
- Optimized **asymmetrical** mooring configurations to **minimize excursions**, control **tensions** and improve **island stability**
- Evaluated **hydrostatic & hydrodynamic** of floating barge using **Maxsurf** and **ANSYS AQWA** for electrical infrastructure
- Investigated **Elastic mooring rope** for floating solar platform under high water-level variations to **improve stability**

Automation & Engineering Tools Development

- Automated **hydrodynamic pressure** extraction from OrcaWave & OrcaFlex using OrcaFlex Python API for FEA analysis
- Developed an **automation tool** to extract properties for steel sections **(IS 811)** using **VBA**, reducing **95%** calculation time
- Created **AUTOLISP** code to automate mooring line coordinates and properties, reducing **70%** time in manual CAD work

Project Execution & Technical Coordination

- Delivered engineering **analysis & design validation** for **8+** floating solar projects from concept development to execution
- Prepared **technical reports** including engineering calculations & simulation documentation for review & client submission
- Collaborated with **multidisciplinary teams** including civil & electrical to integrate simulation insights into optimized design

Installation and Analysis Engineering Intern | TechnipFMC, Hyderabad — **June 2024 - Aug 2024**

- Performed static and dynamic riser simulations using **OrcaFlex** for strength assessment based on Basis of Design (BoD)
- Developed a pre-processing spreadsheet from the load case matrices to automate generation of **96** simulation files (.sim)
- **Automated** post-processing spreadsheet of high volume load cases using **python** code, streamlining HPC Flex extraction
- Packaged Python scripts into executable tool using **PyInstaller**, reducing **90%** manual effort and improving team usability

PROJECTS

Computation Research Project | Master Thesis Project — **Aug 2024 - May 2025**

- Conducted high-fidelity **CFD** simulations on **vortex-induced vibration (VIV)** in flexible marine risers with spanwise groove patterns under the Deep Ocean Mission (DOM) project funded by the Ministry of Earth Sciences (MoES), Govt. of India
- Developed and validated a **FSI** model using **finite element method** within an Arbitrary Lagrangian-Eulerian framework
- Investigated riser suppression through **straight and helical grooves** and quantified reduction up to **40%** in VIV amplitude
- Executed simulations on supercomputer using in-house code, analyzed large-scale simulation data up to **4.2M+ nodes**

Mathematical Model for Prediction of Shallow Water Wave | Bachelor Thesis Project — **Aug 2023 - April 2024**

- Implemented **Physics informed neural network (PINN)** to solve **KdV Burger's** equation to model **shallow water wave** and analyzed wave behavior with and without viscosity
- Applied machine learning techniques like **PINN** and iteratively used **Gradient descent** algorithm for error optimization
- Improved neural network accuracy by reducing **mean square error (MSE)** and validated results with analytical solution

ACHIEVEMENTS, LEADERSHIP & ACTIVITIES

- Recipient of **Institute Silver Medal, Best Master Thesis Project (MTP) award** and Secured **AIR 17** in GATE 2024 Exam
- Secured the **top 3%** in **JEE Main** and **top 5.6%** in **JEE Advanced 2020**, & Recipient **MCM Scholarship** at IIT Kharagpur
- Served as **Secretary, Sports & Games**, Azad Hall, also represented in athletics with multiple medal winning performance
- Appointed as **Teaching Assistant** for Hydrostatics & Stability, and supported the **10+** freshers as mentor under the SWG

EDUCATION

Indian Institute of Technology, Kharagpur — 2020-2025
Dual Degree (B.Tech + M.Tech) in Ocean Engineering & Naval Architecture — GPA: **8.81/10**